

Siddartha Reddy Munugala

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SUMMARY

Software engineer and founder building AI-powered products end to end, from architecture to production deployment. Built Sidewise, ArenaHQ, and Vibeln under Vibeln LLC, reaching 1,321 combined registered users at zero paid acquisition cost. Experienced across frontend, backend, AI infrastructure, deployment pipelines, and cloud services, including on-device and cloud ML inference, encrypted data handling, and production incident response under live user load. Developed multi-agent orchestration tools, audio-AI systems, and automated pipelines, focusing on performance, safety, and rapid product iteration.

SKILLS

Programming: Python, TypeScript, JavaScript, SQL
Frameworks: Next.js 14/15/16, React, Node.js, FastAPI, Express, Tailwind CSS
Databases & Storage: PostgreSQL, Neon Postgres, Drizzle ORM, Supabase, Vercel Blob, Redis
AI & ML: LLMs, RAG, agentic workflows (Claude Agent SDK), TensorFlow.js, Tesseract.js OCR, ElevenLabs API, Gemini API, OpenAI API, LangChain
Tools & Infra: Docker, Google Cloud Run, Vercel, Render, Git, GitHub Actions (CI/CD), WebRTC, WebSockets, REST APIs

EXPERIENCE

Founder and Lead Engineer — Vibeln LLC Jan 2025 - Present
vibeln.us / arenahq.us / sidewise.vercel.app

- Built ArenaHQ (arenahq.us), a live 1v1 competitive coding platform with WebRTC video verification, Supabase Realtime matchmaking, CodeMirror 6, and a sandboxed code execution API, scaling to 800+ users.
- Designed Sidewise's AI research pipeline utilizing Brave Search and Gemini 2.5 Flash, separating retrieval from synthesis to guarantee 100 percent source-backed summaries with zero model-hallucinated citations.
- Launched the flagship anonymous campus social network Vibeln (821 registered users, 40 percent DAU), implementing real-time WebSockets and Supabase row-level security (RLS) policies.
- Diagnosed and resolved 8+ production-critical failures across database and auth pipelines under live user load, maintaining 99 percent or higher uptime.

IT Coordinator and Publicity Chair — NSLS Executive Board Sep 2024 - Present
University of South Dakota

- Automated membership tracking and recruitment workflows using Python and Google Apps Script, reducing onboarding processing time by 50 percent (4 weeks to 2 weeks) and improving reporting accuracy.

PROJECTS

Scout — Real-Time AI Vision Assistant for the Visually Impaired 2026

- Built a three-tier local-first accessibility pipeline (scoutboy.vercel.app) routing between TensorFlow.js COCO-SSD (p50: 80-100ms), Tesseract.js OCR (p50: 200-300ms), and a server-side Gemini 2.5 Flash fallback with a 10s abort ceiling.
- Designed a 100-point frame-differencing canvas filter (12 percent threshold) to skip redundant inferences; offloaded processing to Web Workers and resolved iOS Safari TTS limits via a 20s keepalive and 60s watchdog.

eko — AI-Assisted Multi-Track DAW and Audio Platform 2026 - Present

- Designed a two-service audio platform (Next.js + FastAPI on Cloud Run with non-throttled CPU) performing real-time pitch/rhythm analysis (librosa), MIDI synthesis, ElevenLabs voice cloning, and Demucs stem separation.
- Engineered Discover Mode: a PyTorch autoencoder compressing 70-D audio vectors to 16-D embeddings, integrated with a PennyLane variational quantum circuit (4 qubits) for regression and indexed using Postgres `pgvector` for KNN search.

prism — CTO & Multi-Agent VS Code Extension 2026

- Built a VS Code extension implementing a planner/executor multi-agent system (CTO + 5 role-scoped specialists) wrapping the Claude Agent SDK as an embedded library.
- Engineered a capability-based security model: intercepted and paused all privileged agent tool calls for explicit human approval before filesystem or shell execution.

myera — Autonomous Job-Search Agent and PDF Pipeline 2026

- Built an autonomous cron pipeline (Next.js, Vercel Cron, Claude API, Neon Postgres) that daily sources, classifies, and auto-submits applications with database-backed idempotency.
- Designed a dual-runtime rendering engine using local Puppeteer and serverless [@sparticuz/chromium](https://github.com/sparticuz/chromium) to compile custom HTML resumes to PDF within Vercel's serverless size constraints.

PETRI-Verse — Autonomous Multi-Agent City Simulation 2025

- Developed a persistent, tick-based world simulation engine (Express/Node.js) coordinating autonomous agent governance (proposals/voting), economics, and zoning.
- Engineered a 'Healing Mode' circuit breaker to throttle autonomous actions when stability metrics degrade; streamed real-time state via Server-Sent Events (SSE) across 9 public endpoints.

EDUCATION

B.A. Computer Science, Minor in Innovation and Entrepreneurship Expected May 2027
University of South Dakota

Relevant Coursework: Data Structures and Algorithms, Systems Programming, Database Systems, Software Engineering

LEADERSHIP

- AI and Entrepreneurship Workshop Facilitator: designed and led community sessions for teenagers ages 13-16 covering AI, product building, customer discovery, and public speaking.
- NSLS Executive Board | USD INFORMS (authored 10-week operational plan, built admin portal) | USD Marketing Club